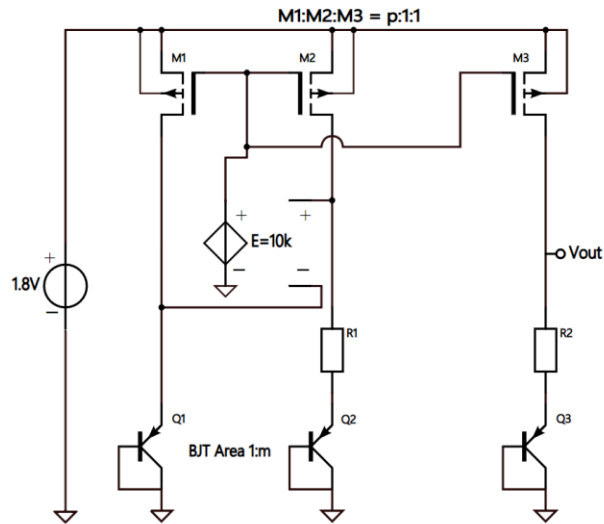




References Assignment - 5714699

a,b)

$$\ln(14) = 2.63906$$

$$V_T = 25.85, \text{ mV}$$

$$\Delta V_{BE} = V_T \ln(14) = 25.85, \text{ mV} \times 2.63906 = 68.22, \text{ mV}$$

$$I = 1, \mu\text{A}$$

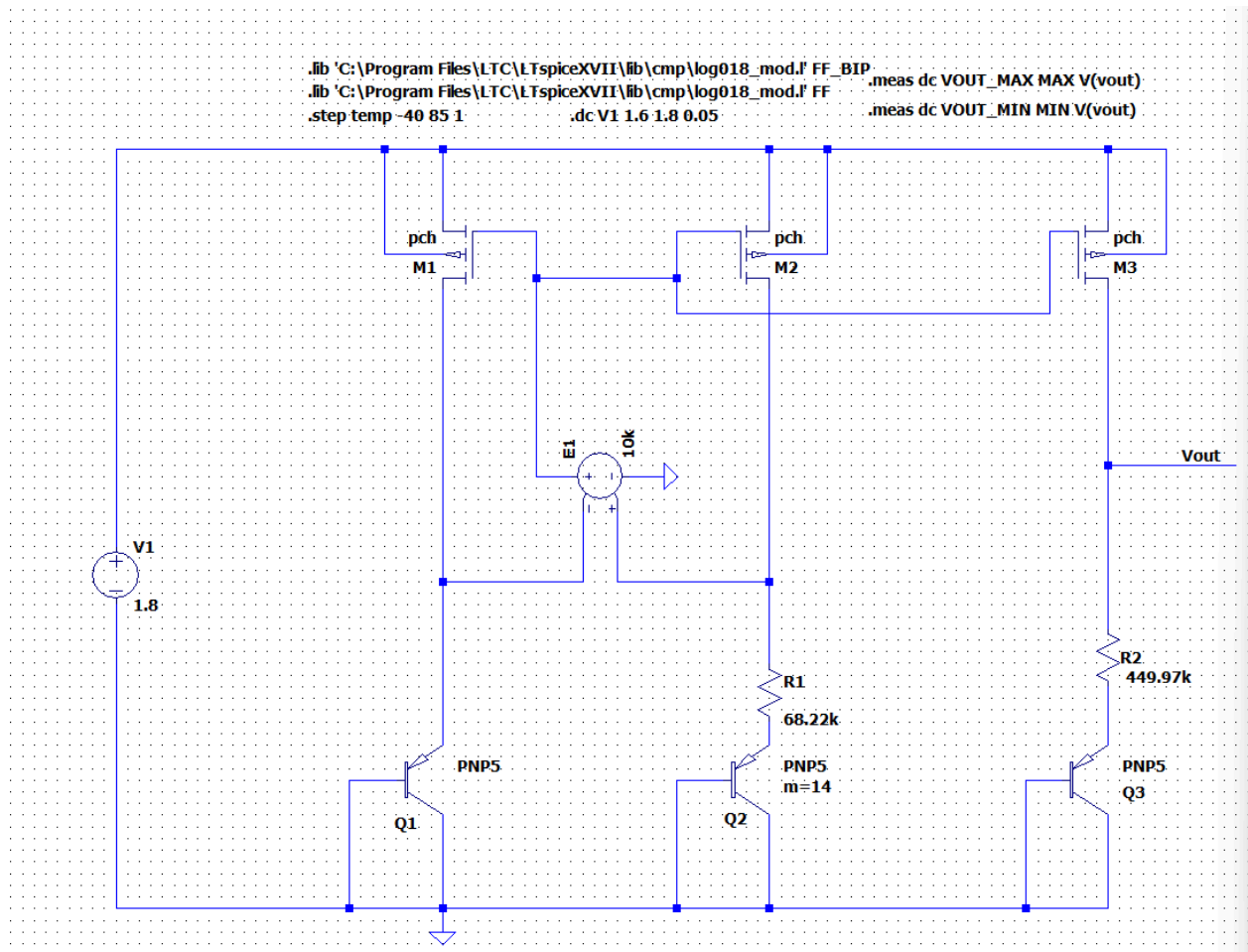
$$R_1 = \frac{\Delta V_{BE}}{I} = \frac{68.22, \text{ mV}}{1, \mu\text{A}} = 68.22, \text{ k}\Omega$$

$$\left(\frac{k}{q}\right) \ln(14) = 0.22742, \text{ mV}/^\circ\text{C}$$

$$\frac{R_2}{R_1} = \frac{1.5}{0.22742} = 6.5958$$

$$R_2 = 6.5958 \times 68.22, \text{ k}\Omega = 449.97, \text{ k}\Omega$$

c)



Monolithic MOSFET - M3

Model Name: OK Cancel

Length(L):

Width(W):

Drain Area(AD):

Source Area(AS):

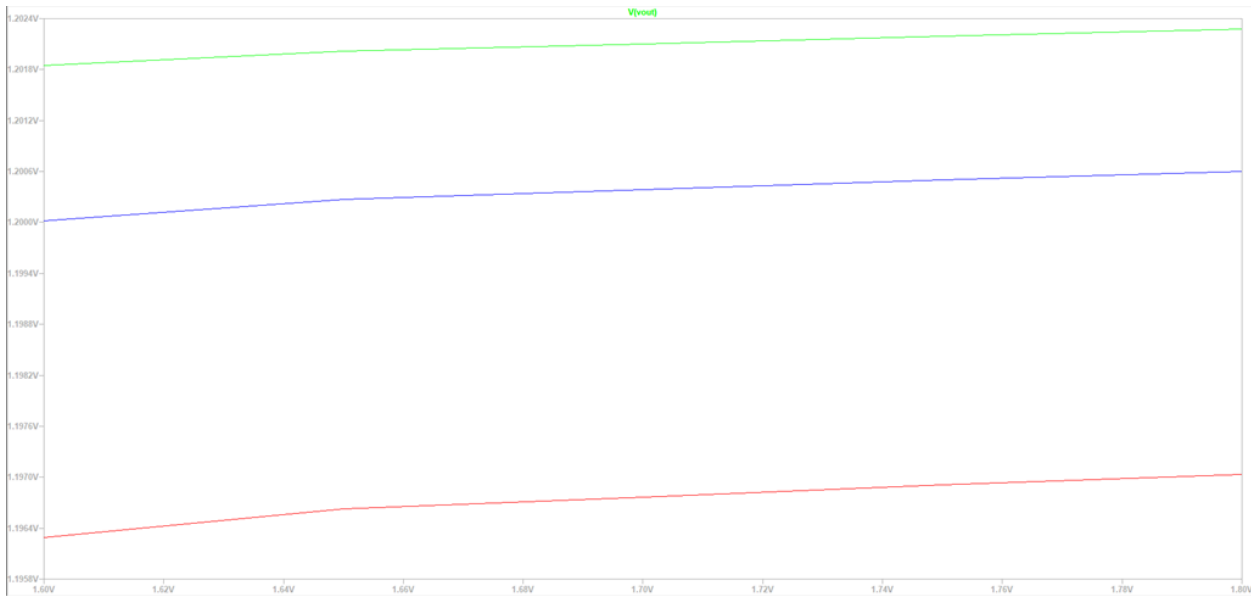
Drain Perimeter(PD):

Source Perimeter(PS):

No. Parallel Devices(M):

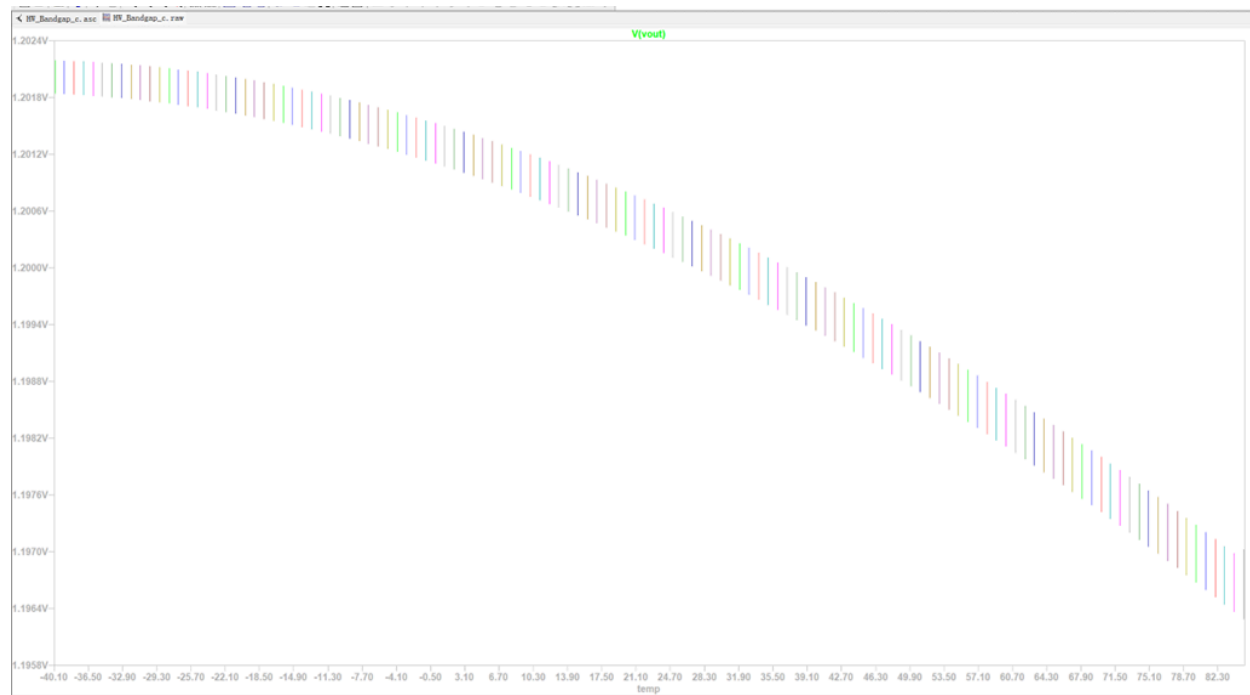
pch l=7u w=20u

TT:

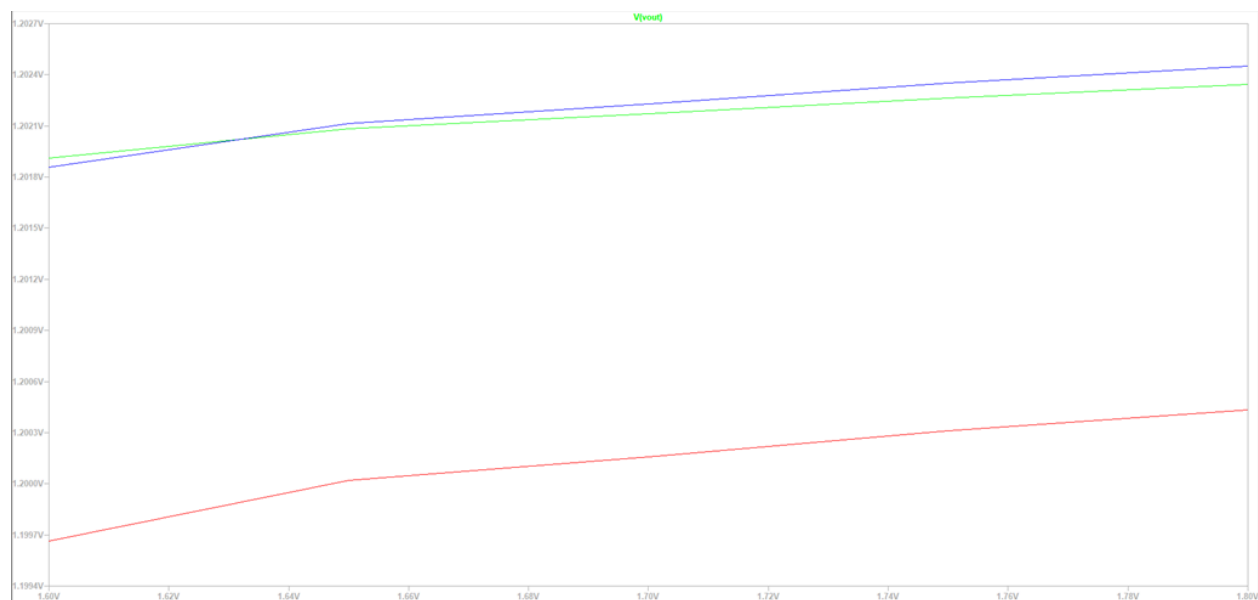


Measurement: vout_max	MAX(v(vout)) [V]	FROM [V]	TO [V]
step 1	1.20231	1.600	1.800
step 2	1.20072	1.600	1.800
step 3	1.19714	1.600	1.800

Measurement: vout_min	MIN(v(vout)) [V]	FROM [V]	TO [V]
step 1	1.20192	1.600	1.800
step 2	1.20011	1.600	1.800
step 3	1.19634	1.600	1.800

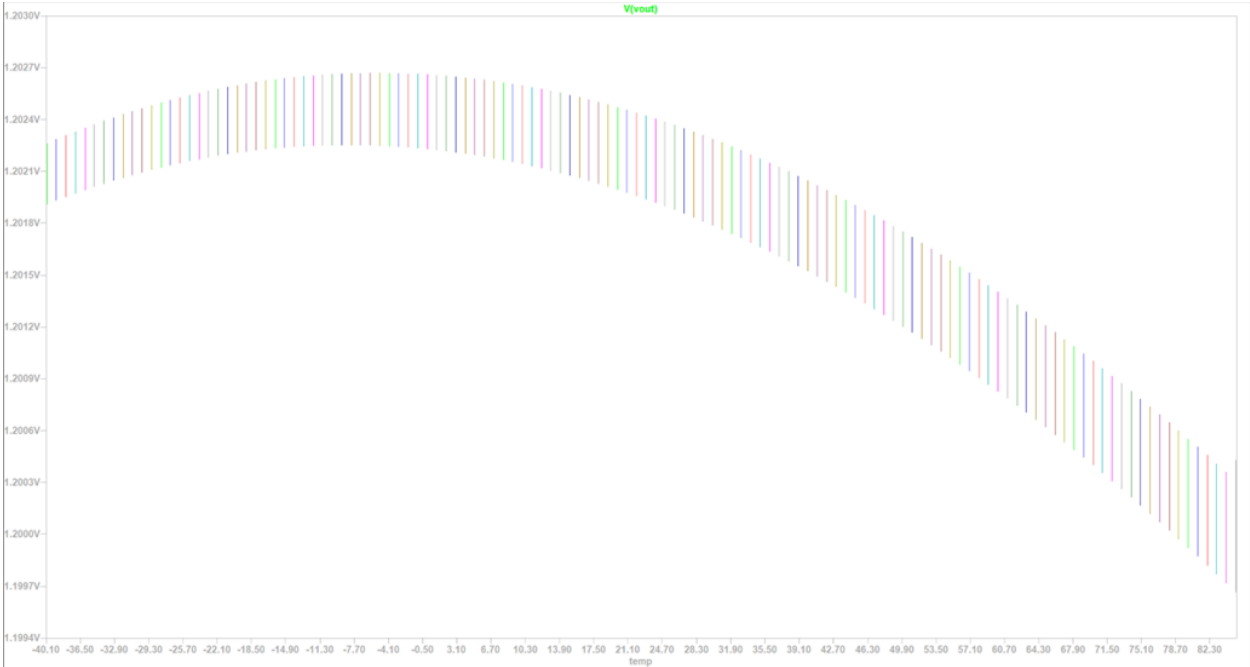


SS:



Measurement: vout_max	MAX(v(vout)) [V]	FROM [V]	TO [V]
step 1	1.20229	1.600	1.800
step 2	1.20256	1.600	1.800
step 3	1.20038	1.600	1.800

Measurement: vout_min	MIN(v(vout)) [V]	FROM [V]	TO [V]
step 1	1.20184	1.600	1.800
step 2	1.20173	1.600	1.800
step 3	1.19961	1.600	1.800



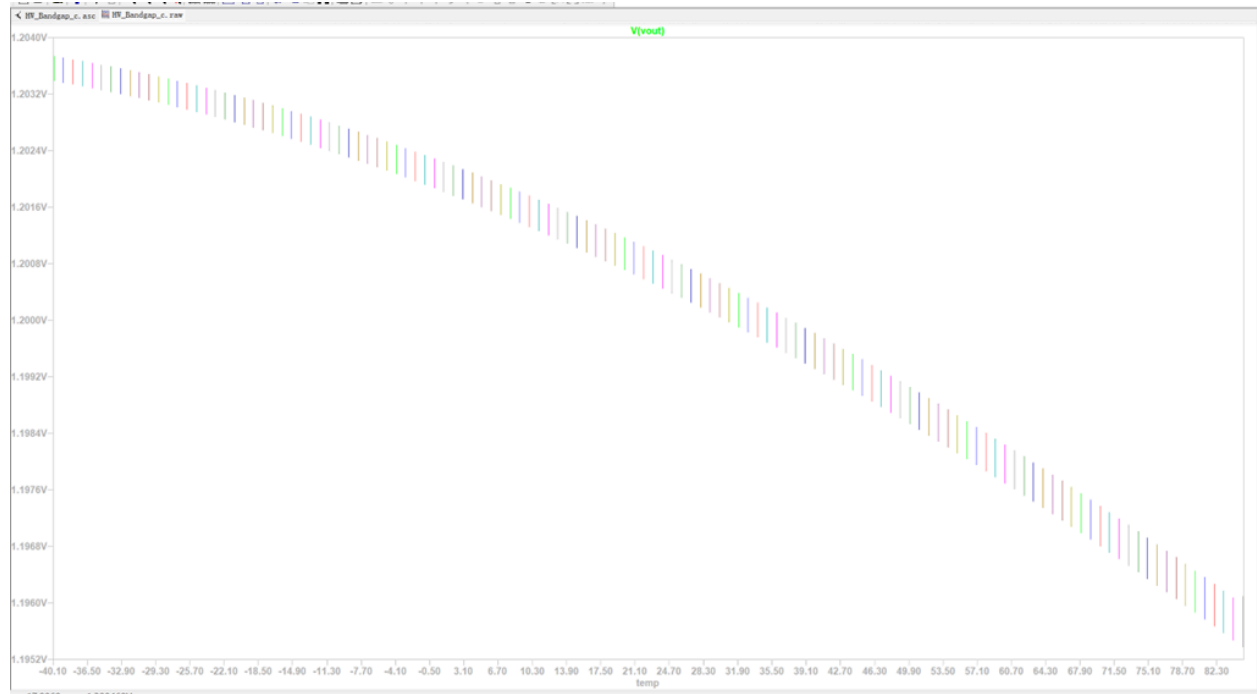
FF:



Measurement: vout_max	MAX(v(vout)) [V]	FROM [V]	TO [V]
step 1	1.20376	1.600	1.800
step 2	1.20102	1.600	1.800
step 3	1.19645	1.600	1.800

Measurement: vout_min	MIN(v(vout)) [V]	FROM [V]	TO [V]
step 1	1.20328	1.600	1.800
step 2	1.20031	1.600	1.800
step 3	1.19544	1.600	1.800

|



$$V_{out_{\max}} = 1.20376 \text{ V}$$

$$V_{out_{\min}} = 1.19544 \text{ V}$$

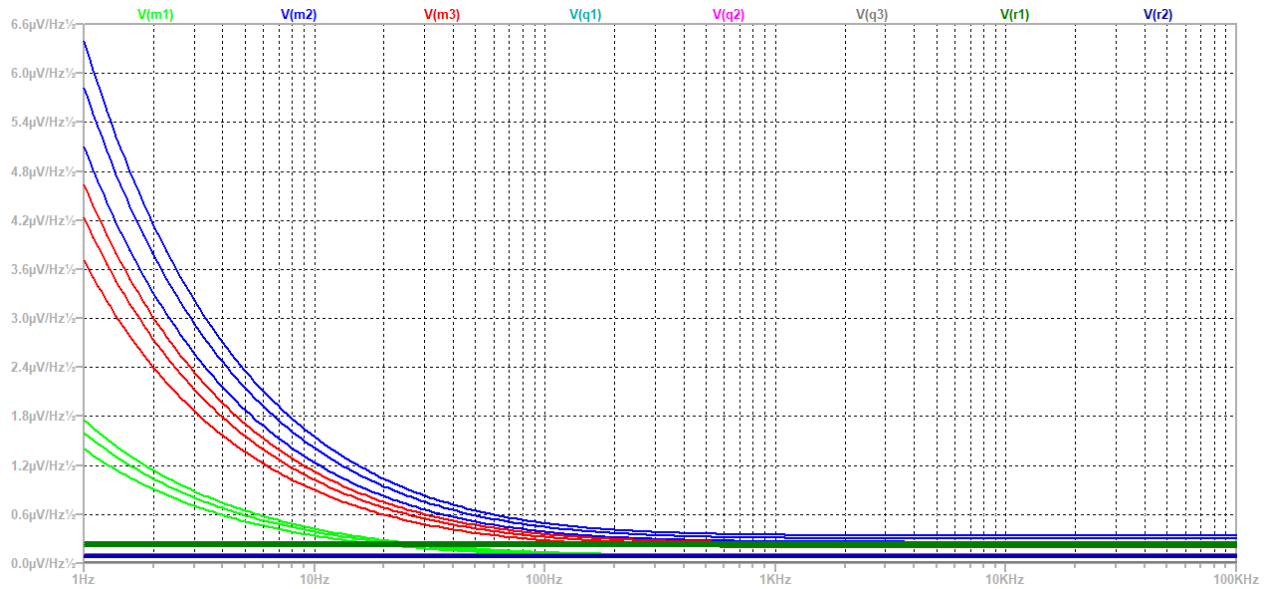
$$V_{out_{\text{mean}}} = 1.200401 \text{ V}$$

$$\text{Spread} = \frac{V_{out_{\max}} - V_{out_{\min}}}{V_{out_{\text{mean}}}} \times 100\%$$

$$\text{Spread} = \frac{1.20376 - 1.19544}{1.200401} \times 100\% = 0.693\%$$

|

d)

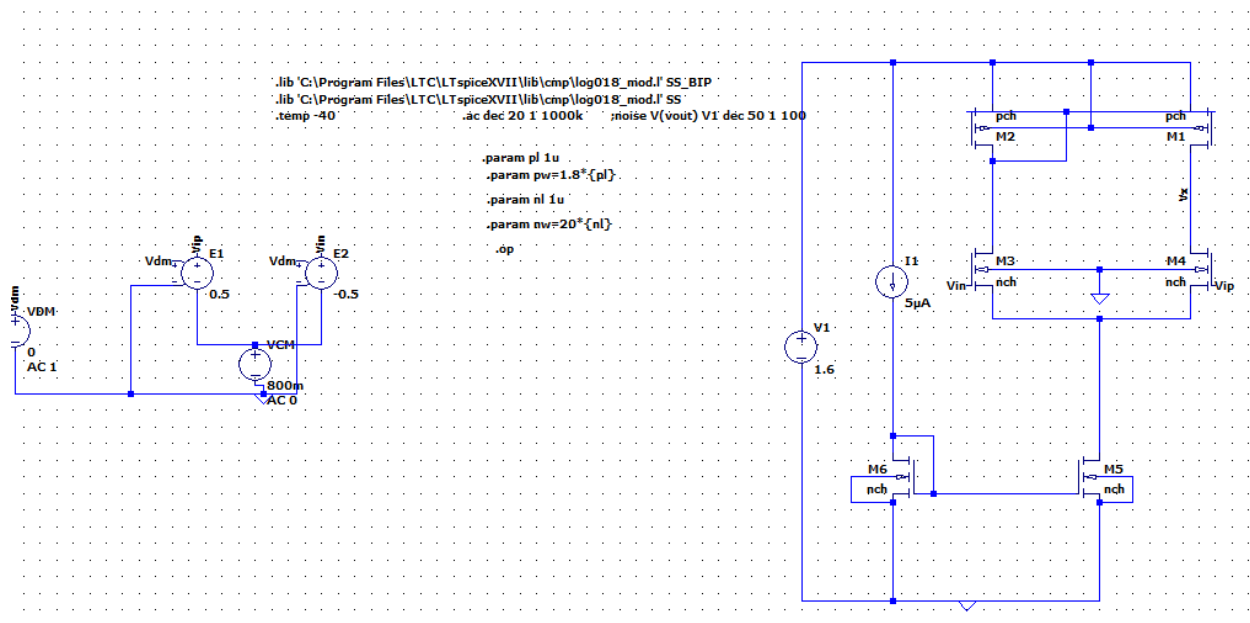


$$V_{out-max} = 1.20562, V; V_{out-min} = 1.19758, V; V_{out-mean} = 1.20210, V; Spread = \frac{V_{out-max} - V_{out-min}}{V_{out-mean}} = 0.669$$

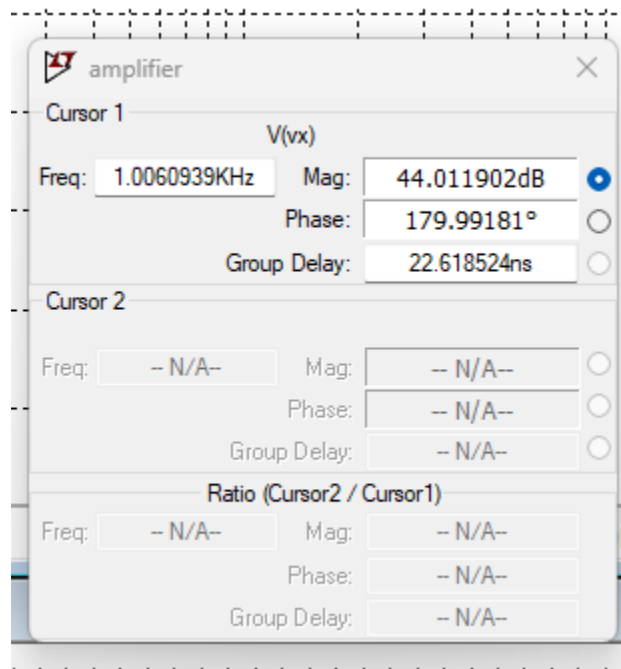
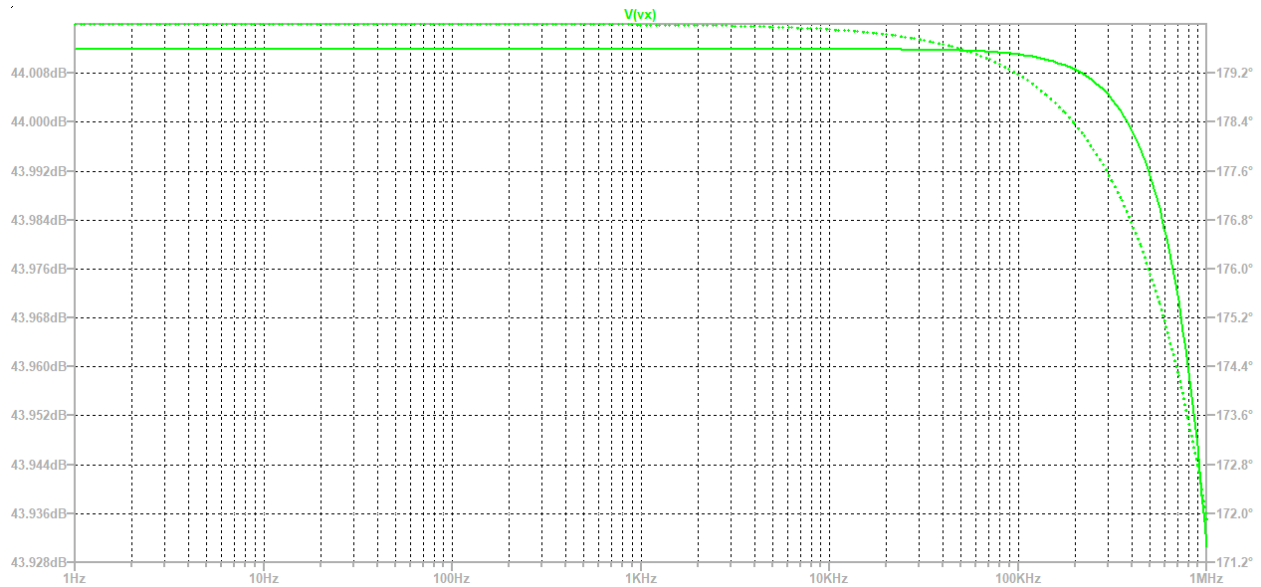
M2 noise is the dominant noise.

e)

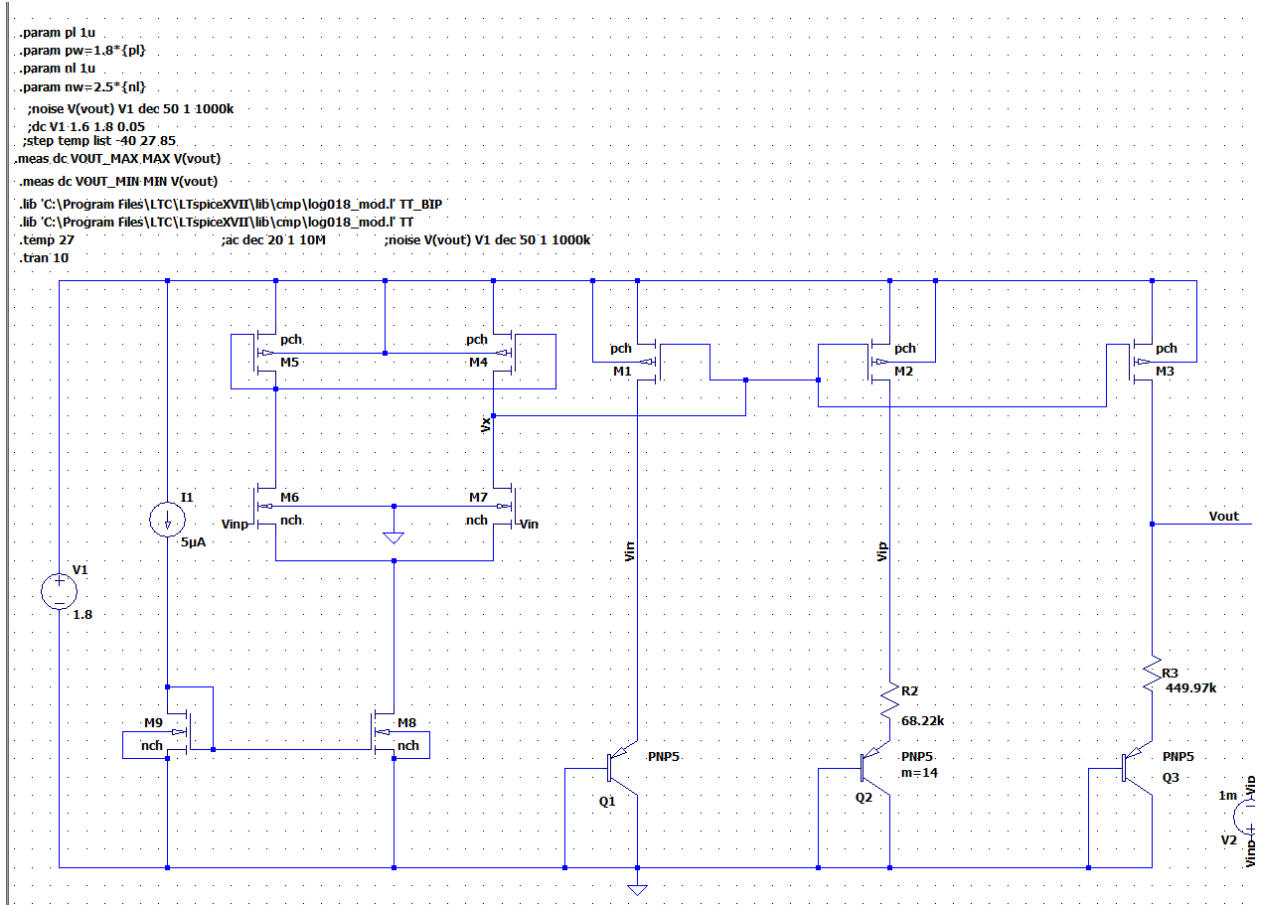
The designed chopper amplifier:



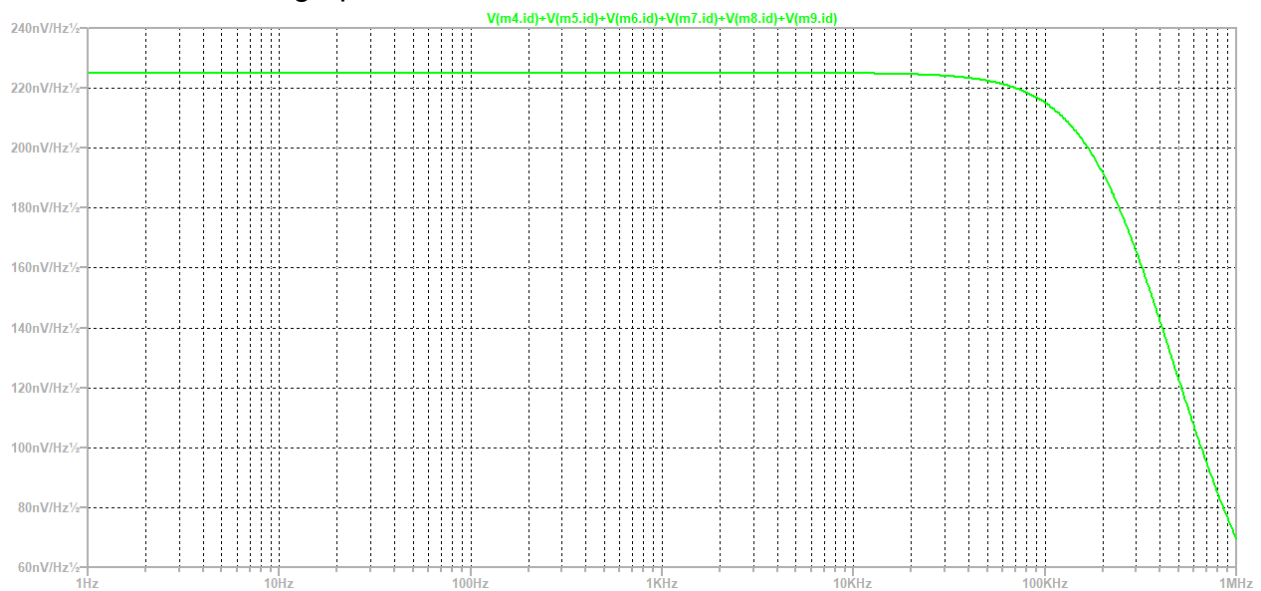
DC gain at SS corner, -40C and 1.6V:

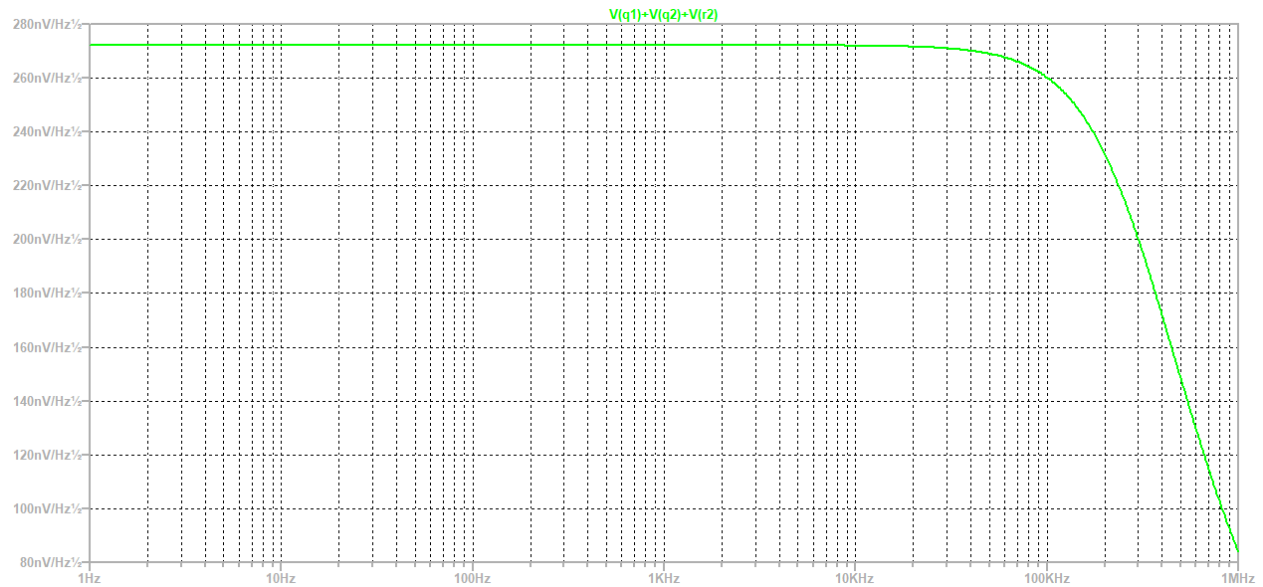


Gain is about 44.5dB which is larger than 40dB.



Noise from the amplifier was observed to be lower than the noise from Q1, Q2 and R1, the below are noise graphs from circuit above:





f)

FF:

Measurement: vout_max

step	MAX(v(vout))	FROM	TO
1	1.72984	1.6	1.8
2	0.974406	1.6	1.8
3	0.53404	1.6	1.8

Measurement: vout_min

step	MIN(v(vout))	FROM	TO
1	1.54752	1.6	1.8
2	0.973809	1.6	1.8
3	0.533891	1.6	1.8

SS:

Measurement: vout_max

step	MAX(v(vout))	FROM	TO
1	1.41654	1.6	1.8
2	0.720074	1.6	1.8
3	0.459113	1.6	1.8

Measurement: vout_min

step	MIN(v(vout))	FROM	TO
1	1.4102	1.6	1.8
2	0.71985	1.6	1.8
3	0.459028	1.6	1.8

TT:

Measurement: vout_max

step	MAX(v(vout))	FROM	TO
1	1.72984	1.6	1.8
2	0.974406	1.6	1.8
3	0.53404	1.6	1.8

Measurement: vout_min

step	MIN(v(vout))	FROM	TO
1	1.54752	1.6	1.8
2	0.973809	1.6	1.8
3	0.533891	1.6	1.8

$$V_{out-max} = 1.72984 \text{ V}$$

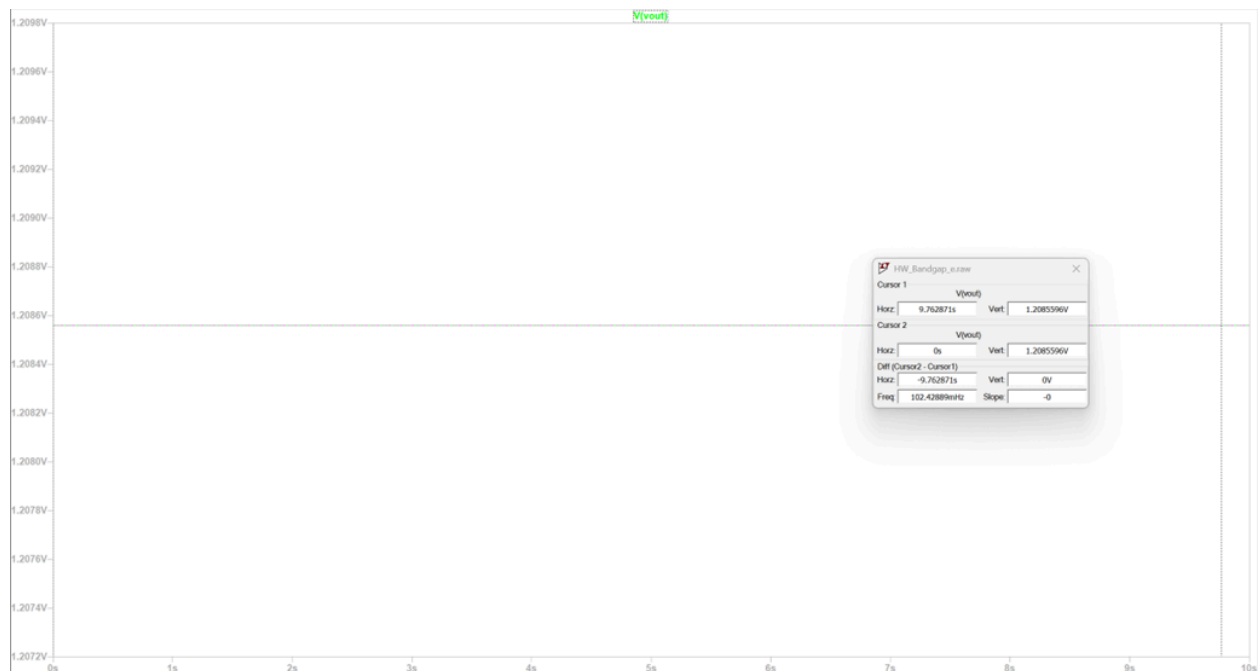
$$V_{out-min} = 0.459028 \text{ V}$$

$$V_{out-mean} = \text{average of six step midpoints} = 0.956526 \text{ V}$$

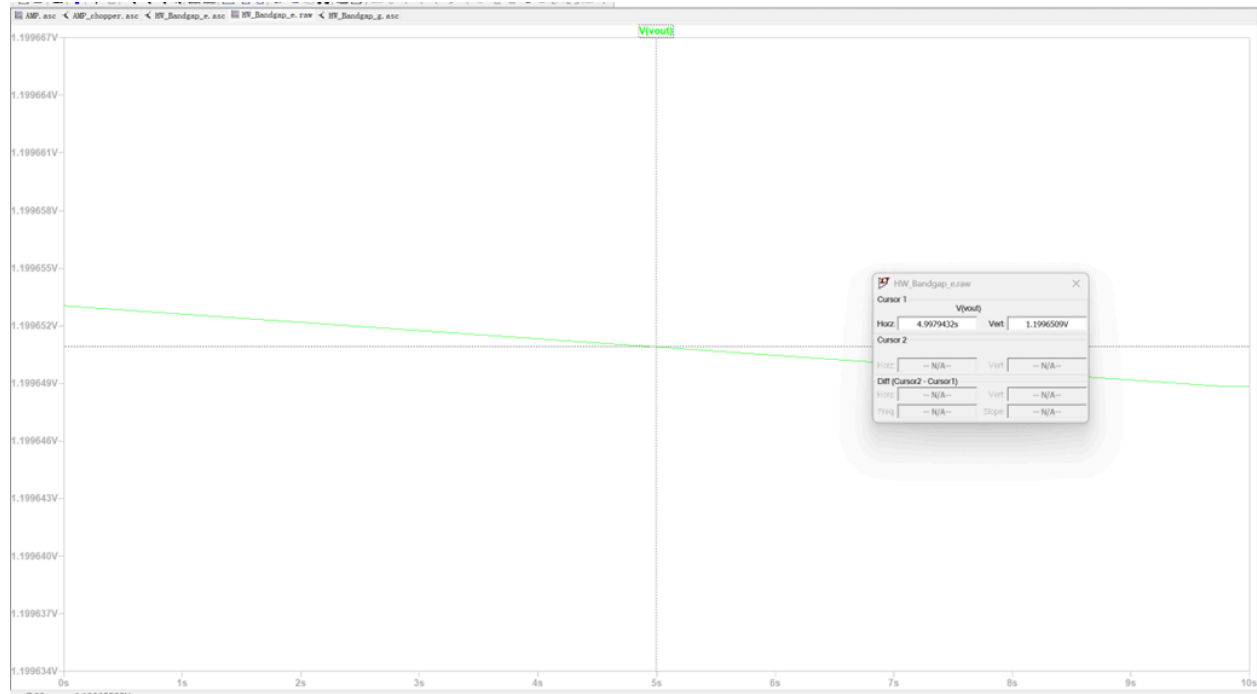
$$\text{Spread} = \frac{V_{out-max} - V_{out-min}}{V_{out-mean}} \times 100\% = \mathbf{132.86\%}$$

g)

Vout at 27C,TT,1.8V:

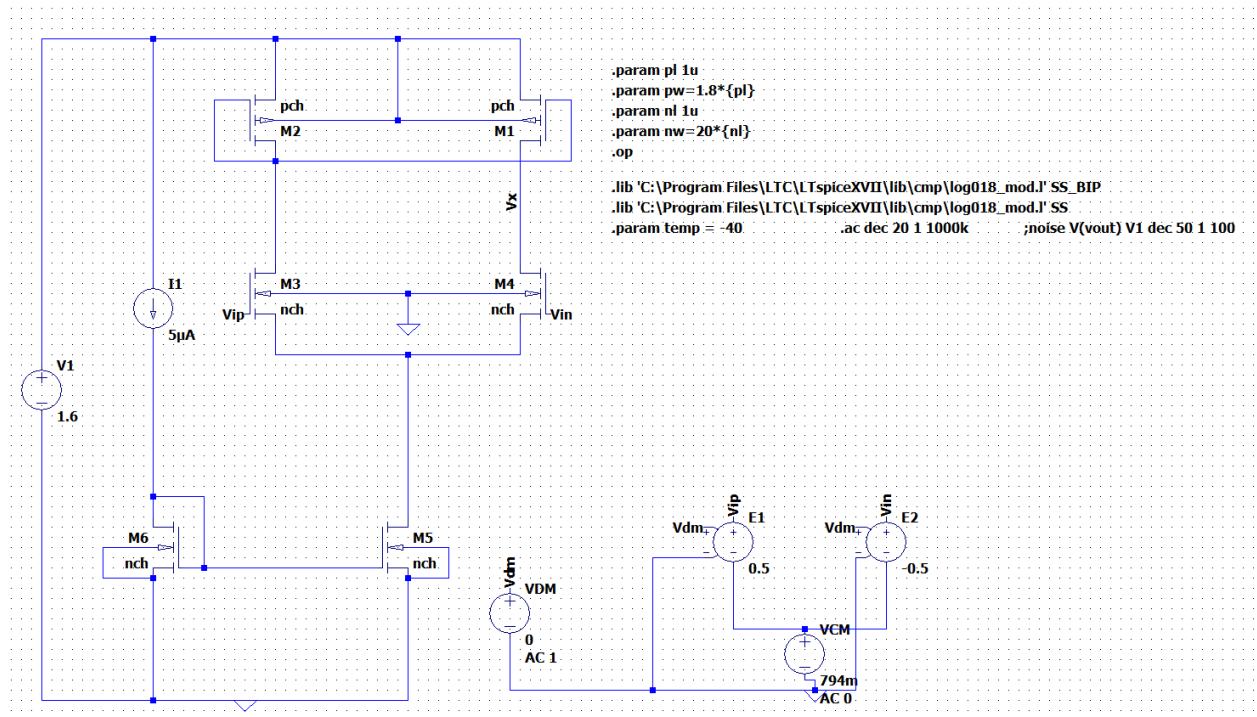


Vout with offset:

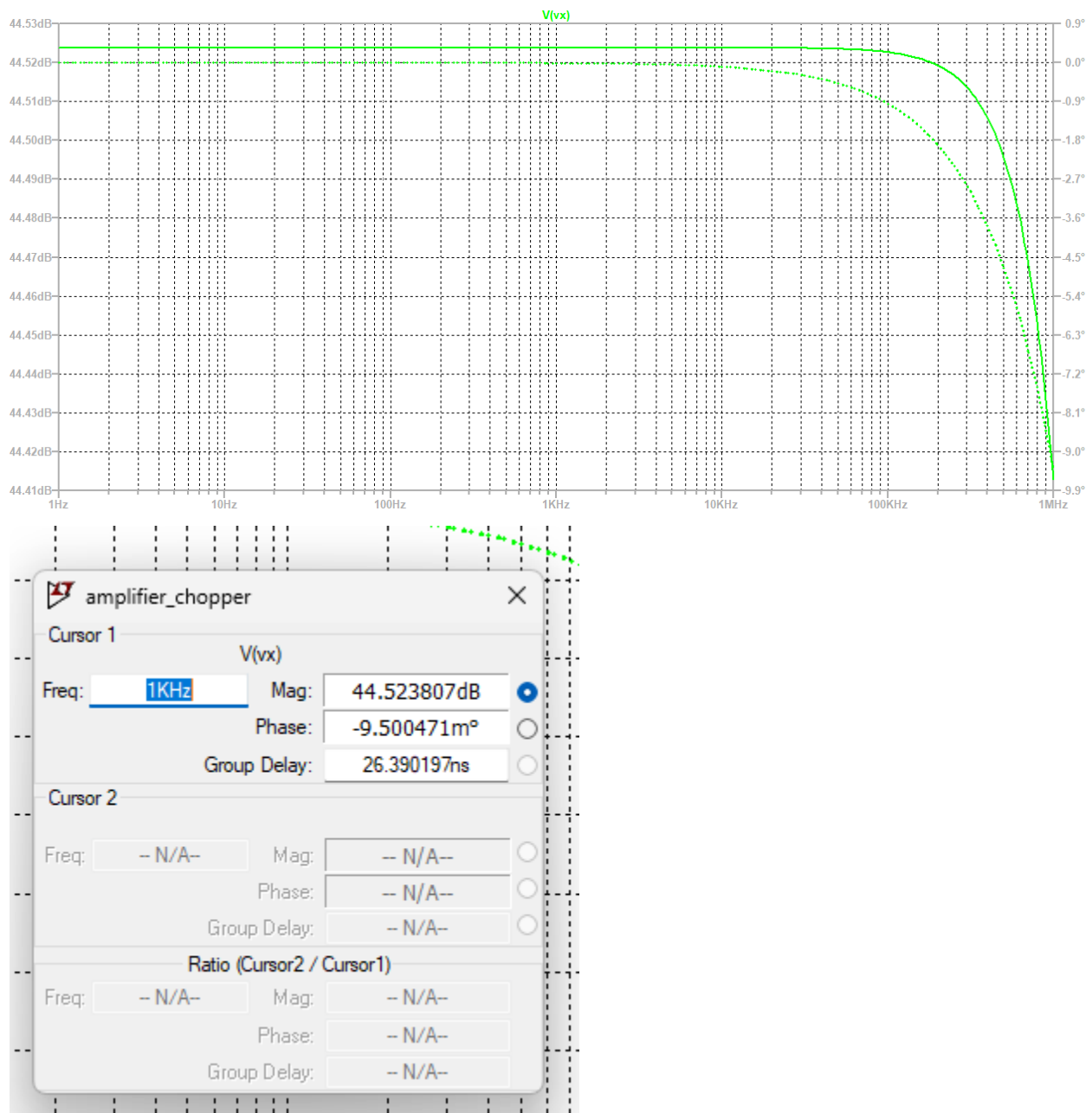


$$\Delta V = 1.20855 - 1.19965 = 8.9, \text{mV}$$

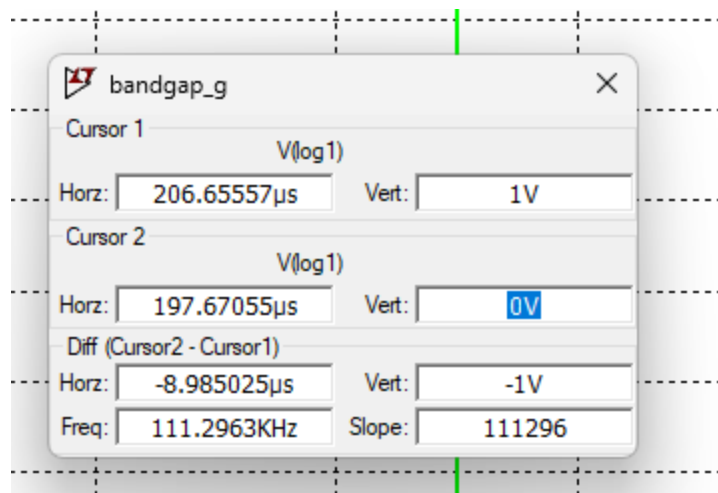
The designed chopper amplifier:



DC gain at SS corner, -40C and 1.6V:



Chopping circuit:



Mean output voltage: $V_{out,avg} = 1.2088, V$

Residual offset: $|V_{out,avg} - V_{out,measured}| = |1.2088 - 1.20855| = 0.25, mV$

Therefore, this operating frequency results in a very small output error.